

Modeling the Joint Distribution of Firm Characteristics and Missingness with Autoregressive Density Estimation

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1. Motivation and Research Question

Empirical asset-pricing datasets often contain substantial missingness across firm characteristics. In many applications, missing observations are handled through deletion or simple imputation before the data are used in return-prediction or asset-pricing models. However, missingness itself may contain systematic information: whether one characteristic is missing may be related to the observed values or missingness of other characteristics.

This project studies this dependence by estimating the *joint distribution* of firm characteristics and their missingness patterns. Rather than predicting a single outcome variable, the primary statistical object is

$$p_{\theta}(B_{it}, M_{it}),$$

where B_{it} represents the binary positions of firm i 's observed characteristics in quarter t , and M_{it} contains the corresponding missingness indicators. The main question is whether a flexible tractable-likelihood model can capture systematic dependence both across firm characteristics and across their missingness patterns.

2. Data and Binary Representation

The dataset is a processed firm-quarter panel containing 809,472 observations for 21,227 publicly traded securities from 1984 to 2020. It contains 45 firm-level characteristics related to valuation,

profitability, investment, momentum, trading, and risk. The characteristics have already been transformed into within-quarter cross-sectional ranks lying approximately in $[-0.5, 0.5]$. A separate dataset provides eight quarterly macroeconomic and market variables that can be matched to the firm panel over the full sample period.

Missingness is substantial and heterogeneous across both firms and characteristics. For each characteristic j , I define

$$M_{itj} = \mathbf{1}\{X_{itj} \text{ is missing}\},$$

and, when the characteristic is observed,

$$B_{itj} = \mathbf{1}\{X_{itj} > 0\}.$$

Since zero approximately represents the cross-sectional median rank, B_{itj} indicates whether a firm lies in the upper or lower half of the cross-sectional distribution. The pair (B_{itj}, M_{itj}) therefore distinguishes three states: observed below the median, observed above the median, and missing. Stacking these variables produces a high-dimensional binary representation of each firm-quarter observation.

3. Autoregressive Likelihood Estimation

Let

$$Y_{it} = (B_{it}, M_{it}) \in \{0, 1\}^D.$$

Directly assigning probabilities to all possible realizations of Y_{it} is infeasible because the size of the state space grows exponentially with D . I therefore plan to use the autoregressive factorization

$$p_{\theta}(Y_{it}) = \prod_{d=1}^D p_{\theta}(Y_{it,d} \mid Y_{it,1}, \dots, Y_{it,d-1}).$$

Because each component is binary, the conditional distributions are Bernoulli, allowing the full joint likelihood to remain exactly normalized and tractable.

I will begin with an independent Bernoulli model and a simple autoregressive logistic specification as benchmarks. I will then estimate more flexible neural autoregressive models, particularly NADE and MADE. NADE shares neural representations across the sequence of conditional distributions, while MADE uses masked connections in a feed-forward neural network to ensure that each output depends only on preceding variables. These models allow nonlinear dependence among characteristics and missingness indicators while retaining an exactly evaluable likelihood.

4. Evaluation and Extensions

The primary evaluation criterion will be held-out negative log likelihood,

$$\text{NLL} = -\frac{1}{N_{\text{test}}} \sum_{(i,t) \in \text{test}} \log p_{\hat{\theta}}(Y_{it}).$$

Comparing the independent Bernoulli, autoregressive logistic, and MADE/NADE models will show how much likelihood improvement is obtained from modeling dependence across characteristics and missingness indicators.

Beyond overall likelihood performance, I will examine conditional distributions such as

$$p_{\theta}(M_{itj} = 1 \mid B_{it,-j}, M_{it,-j})$$

to study which observed firm characteristics and missingness configurations are most strongly associated with missingness in characteristic j . Possible extensions include conditioning the joint distribution on quarterly macroeconomic variables,

$$p_{\theta}(B_{it}, M_{it} \mid Z_t),$$

and investigating whether the learned dependence structure can improve imputation or downstream asset-pricing applications.